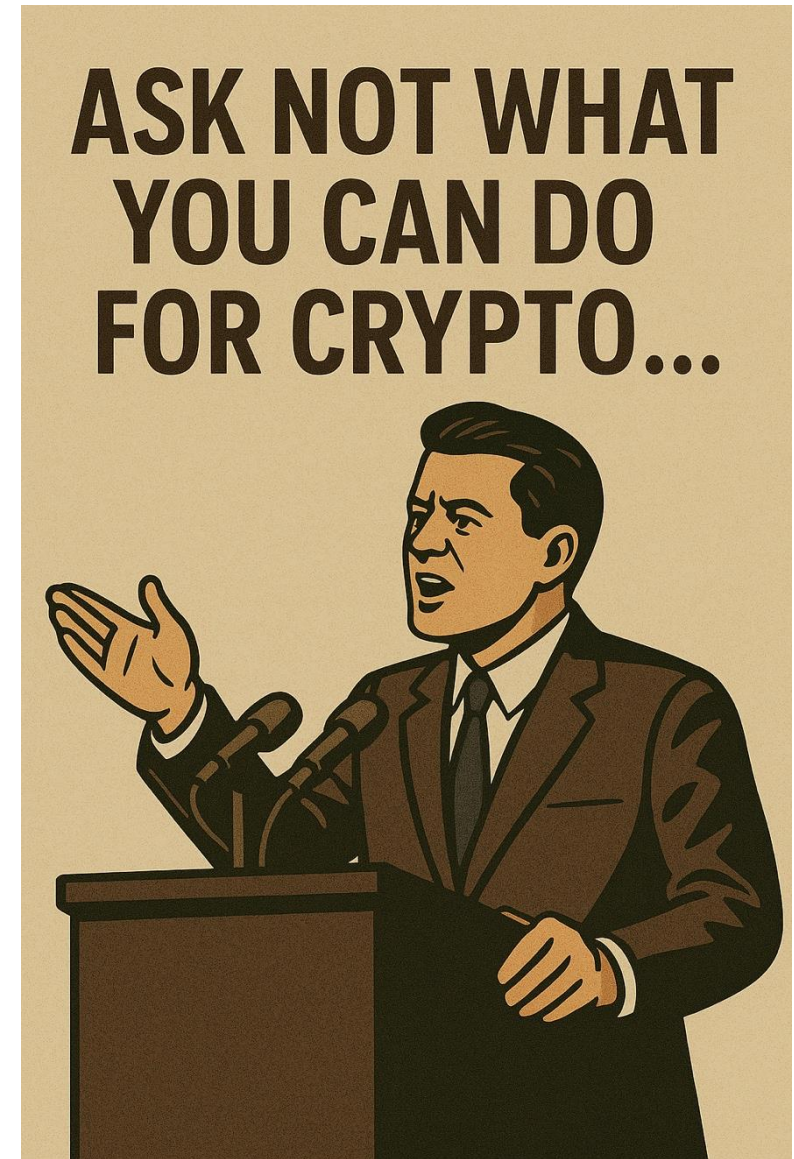


Cryptography and AI: Challenges and Opportunities

Vinod Vaikuntanathan

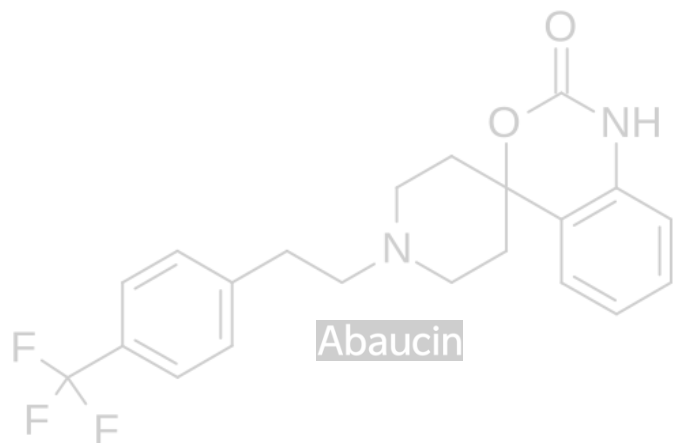


Based on joint works with **Andrej Bogdanov, Elette Boyle, Rio Lavigne, Alon Rosen, Neekon Vafa, and Or Zamir**. Some slides are borrowed from these generous people.

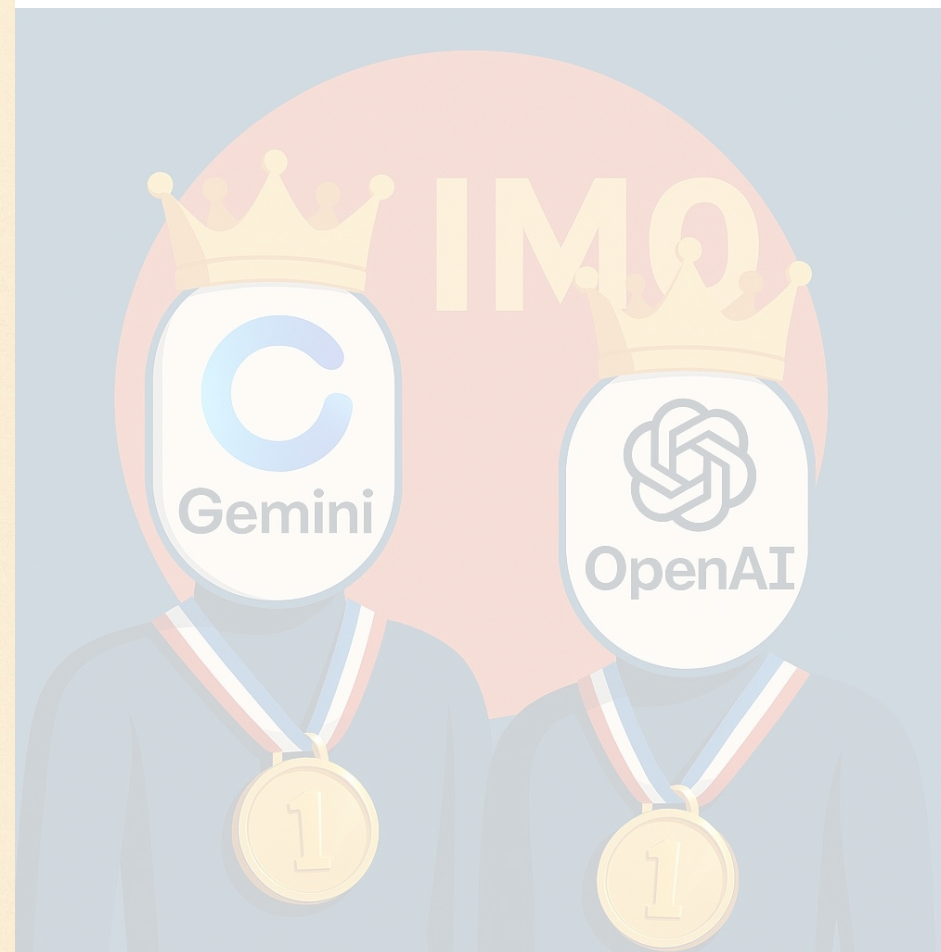
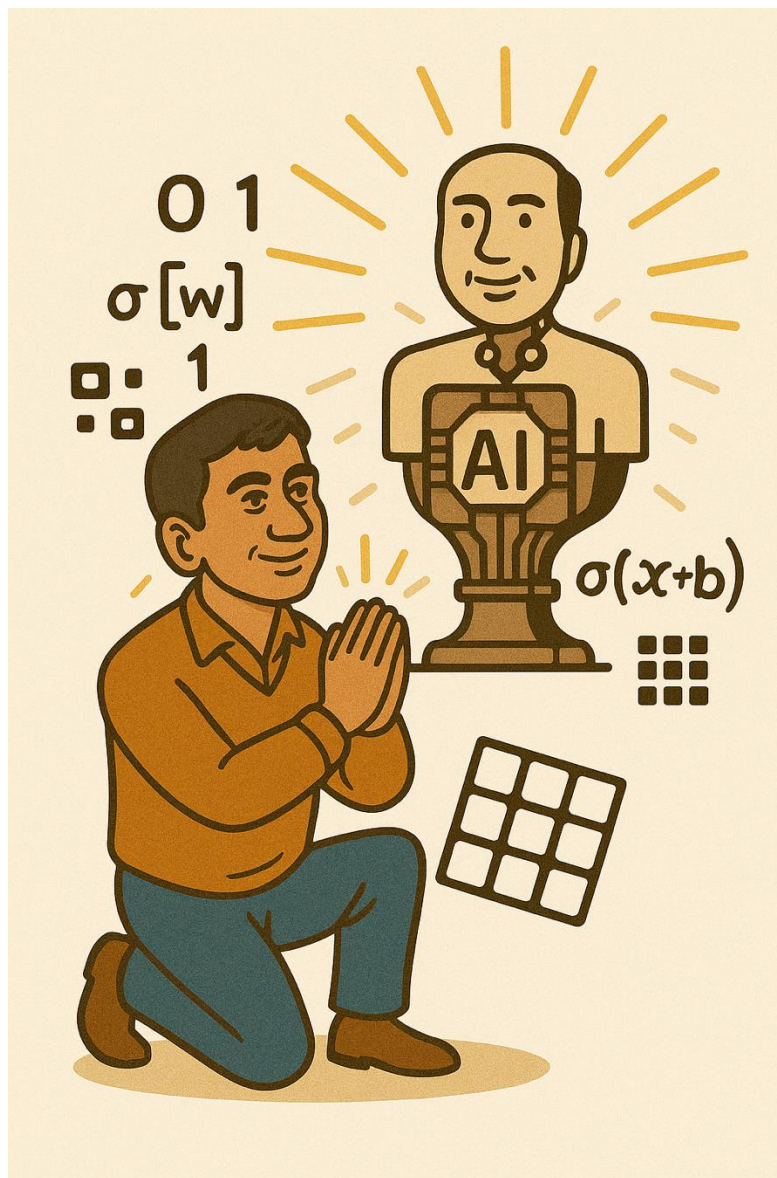


(with apologies to Or Zamir)

AI is here...



Scientific Discovery



Mathematical Problem Solving



https://en.wikipedia.org/wiki/Stochastic_parrot



<https://ai-2027.com>

(Some) AI Challenges

Trustworthy AI

- ◆ **Hallucination:** *how do I know the model outputs are correct?*
 - ◆ **Robustness:** *will the model remain good under distribution shift?*
 - ◆ **(Adversarial) Control:** *susceptible to undetectable manipulation?*
 - ◆ **Alignment:** *are the model's goals aligned to ours?*
-
- ◆ **Resource Consumption:** *can we lower the giant compute use?*

My Thesis:

Cryptography and Cryptographers have a Role to Play

◆ Adversarial Thinking

The Omnipresent Adversary



◆ Definitions cf. Naor.

What access does the adversary have?

What are her goals?

What measures her success?

◆ Constructions: Turning Hardness into Usefulness



Hard Problems



Useful Systems

◆ Proofs via Reductions (or the “win-win paradigm”):



CAPTCHA:
Using Hard AI Problems For Security

Luis von Ahn¹, Manuel Blum¹, Nicholas J. Hopper¹, and John Langford²

¹ Computer Science Dept., Carnegie Mellon University, Pittsburgh PA 15213, USA

² IBM T.J. Watson Research Center, Yorktown Heights NY 10598, USA

Verification of a human in the loop
or
Identification via the Turing Test*

Moni Naor [†]

September 13th, 1996

... but we need to think differently

Cryptography

theory first
adversarial
maximalistic



ML/AI

empirical
optimistic
pragmatic

Different models, Different goals, Different adversaries.

Need new ideas, new tools, new hard problems.

This Talk: Two Vignettes

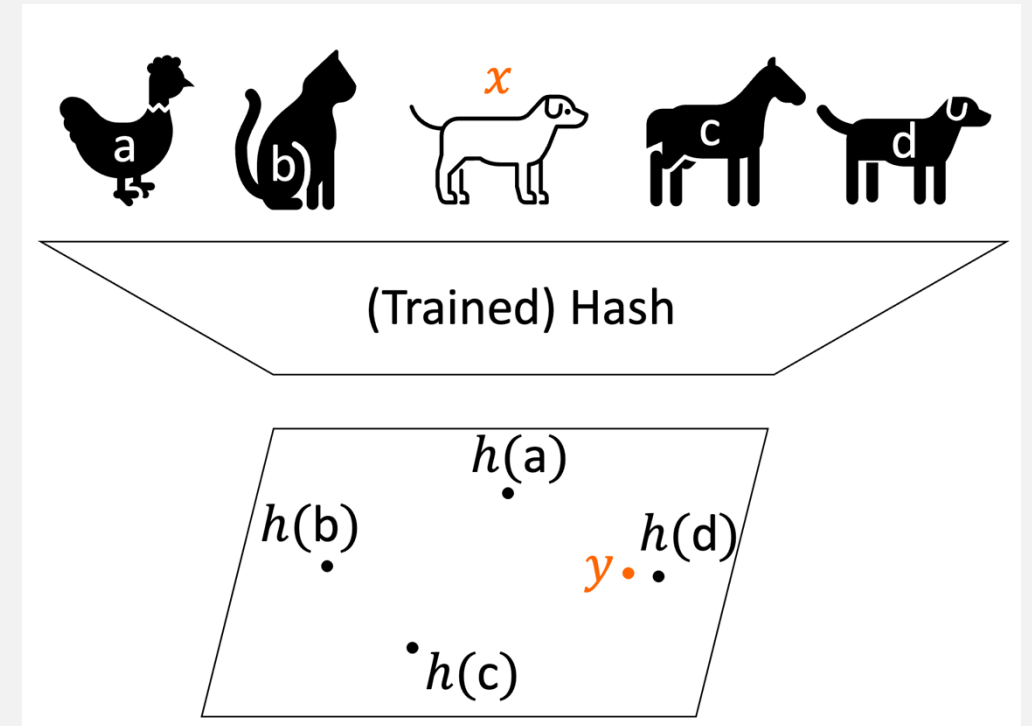
Robust Embeddings

(the notion of robust locality-sensitive hash functions)

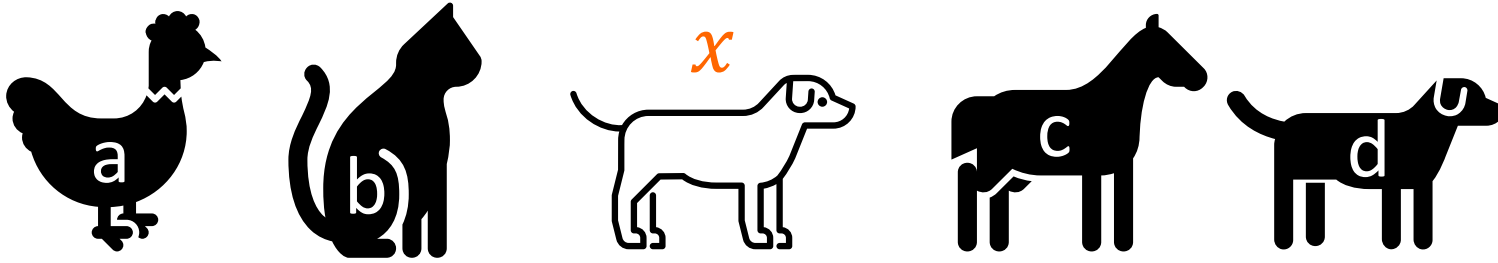
Crypto to Speed up (ML) Algorithms

(the notion of trapdoored matrices)

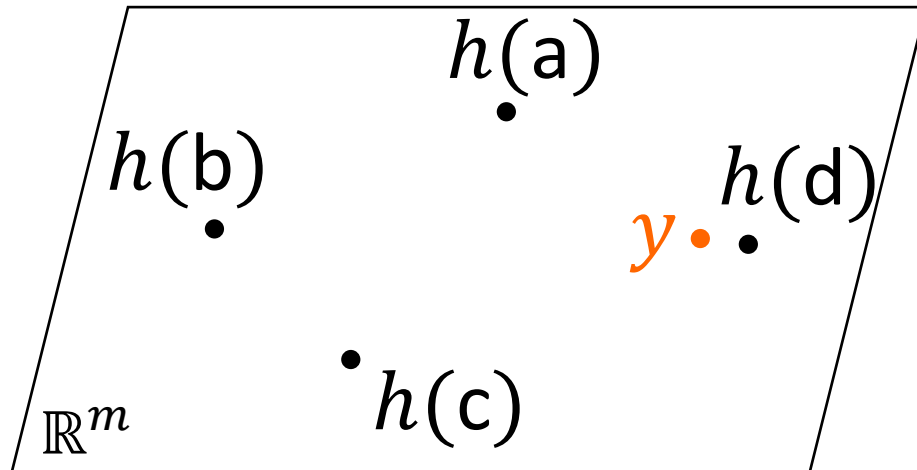
Robust ML Embeddings



ML Embeddings



(Trained) Hash



Semantic Similarity

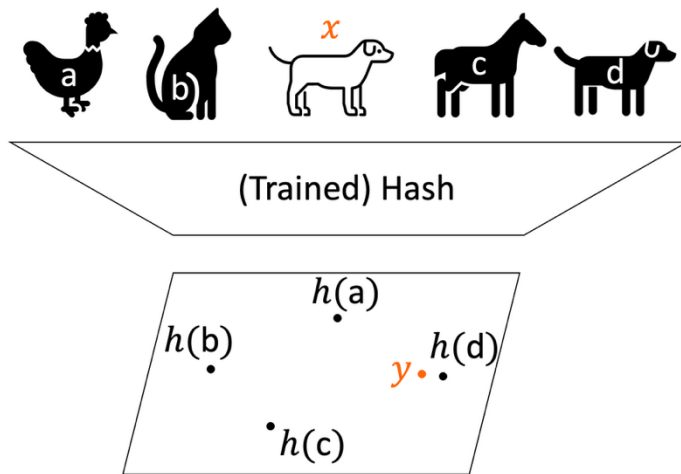


Geometric Closeness

ML Embeddings

Word2Vec [Mikolov et al. 2013]

Early neural system that mapped words to vectors.



Vision Transformers [Dosovitsky et al. 2021]

Compressing map from e.g. 224 X 224 RGB image to a 768-dimensional vector (with 32-bit precision)

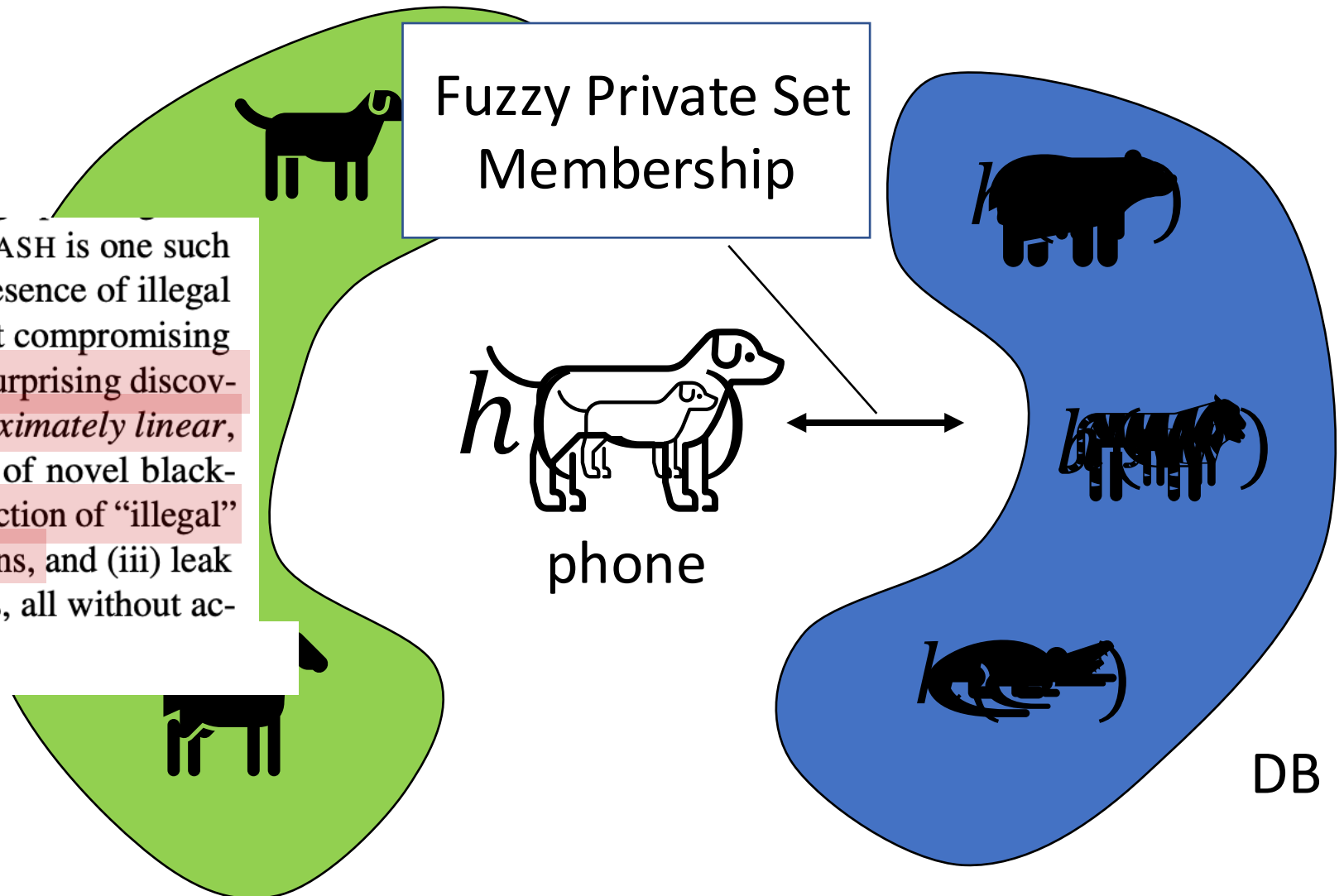
CLIP [Li et al. 2016, Radford et al. 2021]

Multimodal: connect text and image embeddings!

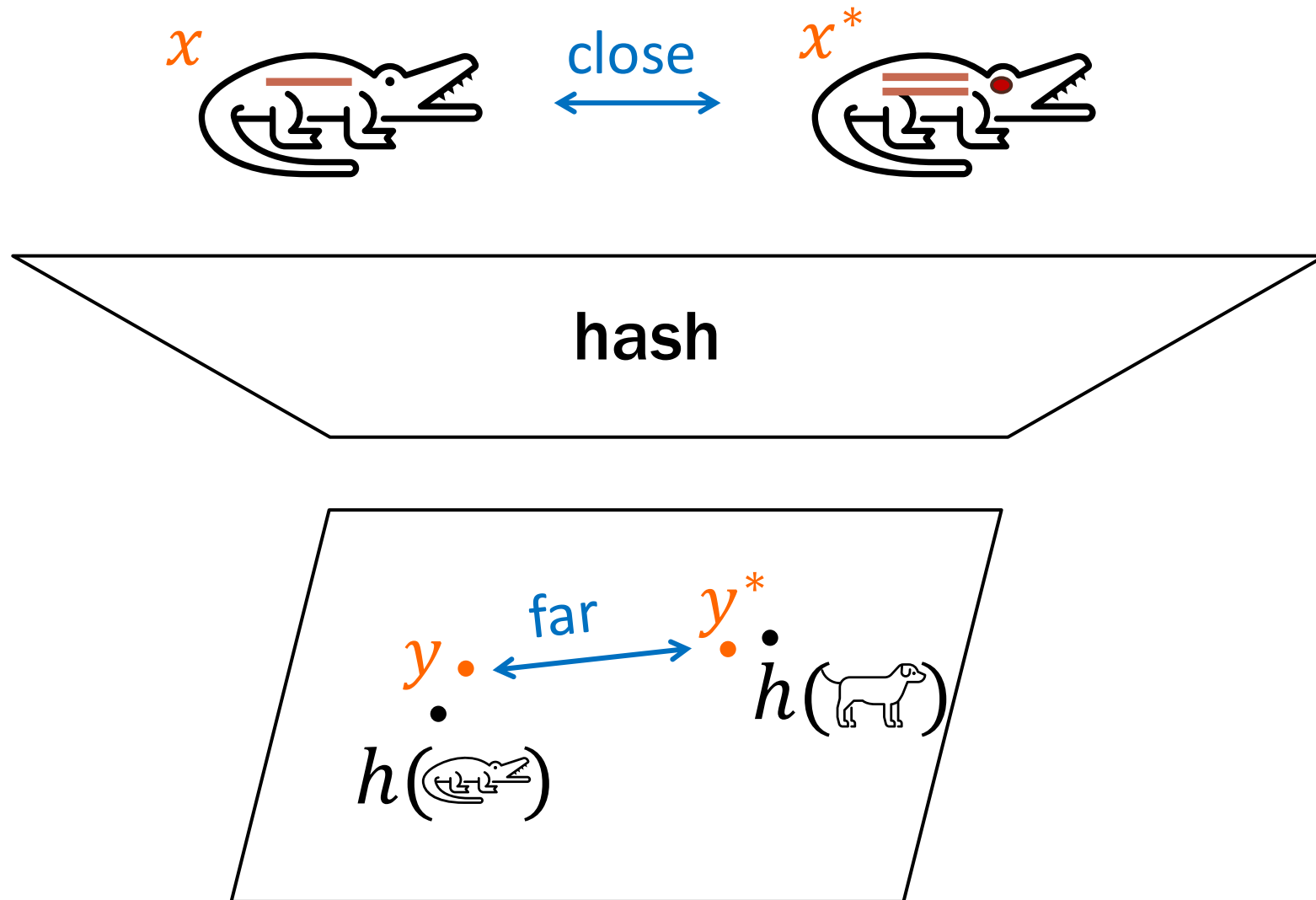
Apple NeuralHash [2021]

[Bhatia, Meng '22]

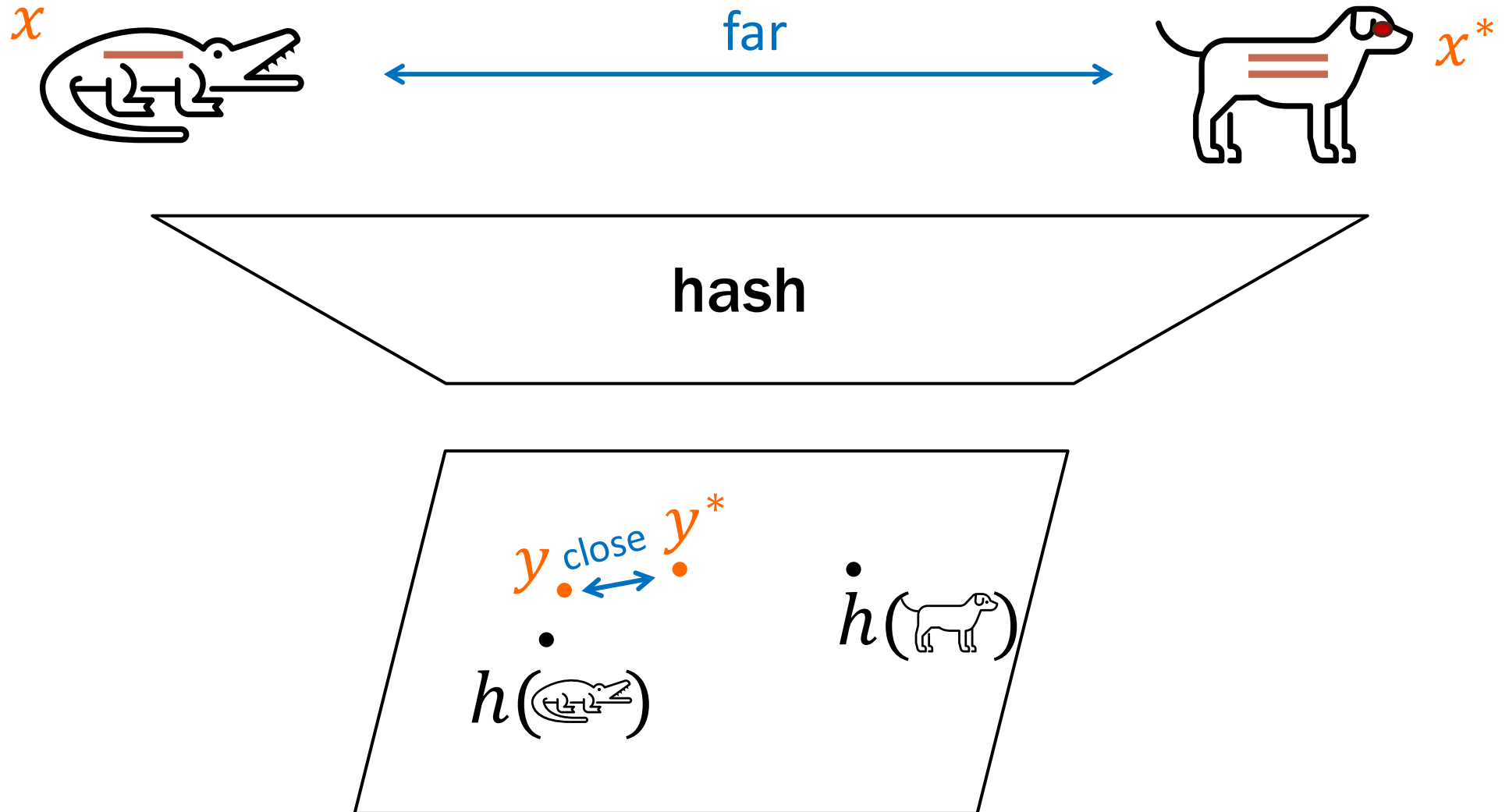
Apple's NEURALHASH is one such system that aims to detect the presence of illegal content on users' devices without compromising consumer privacy. We make the surprising discovery that NEURALHASH is *approximately linear*, which inspires the development of novel black-box attacks that can (i) evade detection of "illegal" images, (ii) generate near-collisions, and (iii) leak information about hashed images, all without access to model parameters.



Adversarial Expansion (Evasion)



Adversarial Contraction (Framing)





Robust Locality-Sensitive Hash

[Boyle-Lavigne-**V.**'19]

A hash function family $\{h_K: \mathbb{R}^n \rightarrow \mathbb{R}^m\}$ is a robust LSH (for a given norm) if:

- **Compressing**: The output length (in bits) is smaller than the input length.
- **α -expanding (for $\alpha > 1$)**: given K , no p.p.t. adversary can find $x, y \in \mathbb{R}^n$ s.t. $\|h_K(x) - h_K(y)\| > \alpha \cdot \|x - y\|$.
- **β -contracting (for $\beta \leq 1$)**: given K , no p.p.t. adversary can find $x, y \in \mathbb{R}^n$ s.t. $\|h_K(x) - h_K(y)\| < \beta \cdot \|x - y\|$.

(Computational) Distortion: α/β , ideally close to 1.

Example: Robust Hash for Equality

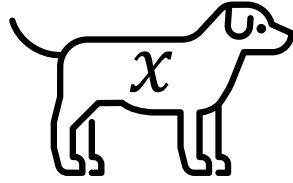
WANT: When $x = y$, $h(x) = h(y)$. (trivial)

When $x \neq y$, no algorithm can ensure that $h(x) = h(y)$.

Key Question: Does the algorithm (adversary) know h ?

NO: universal hash functions (from TCS)

YES: collision-resistant hash functions (from cryptography)



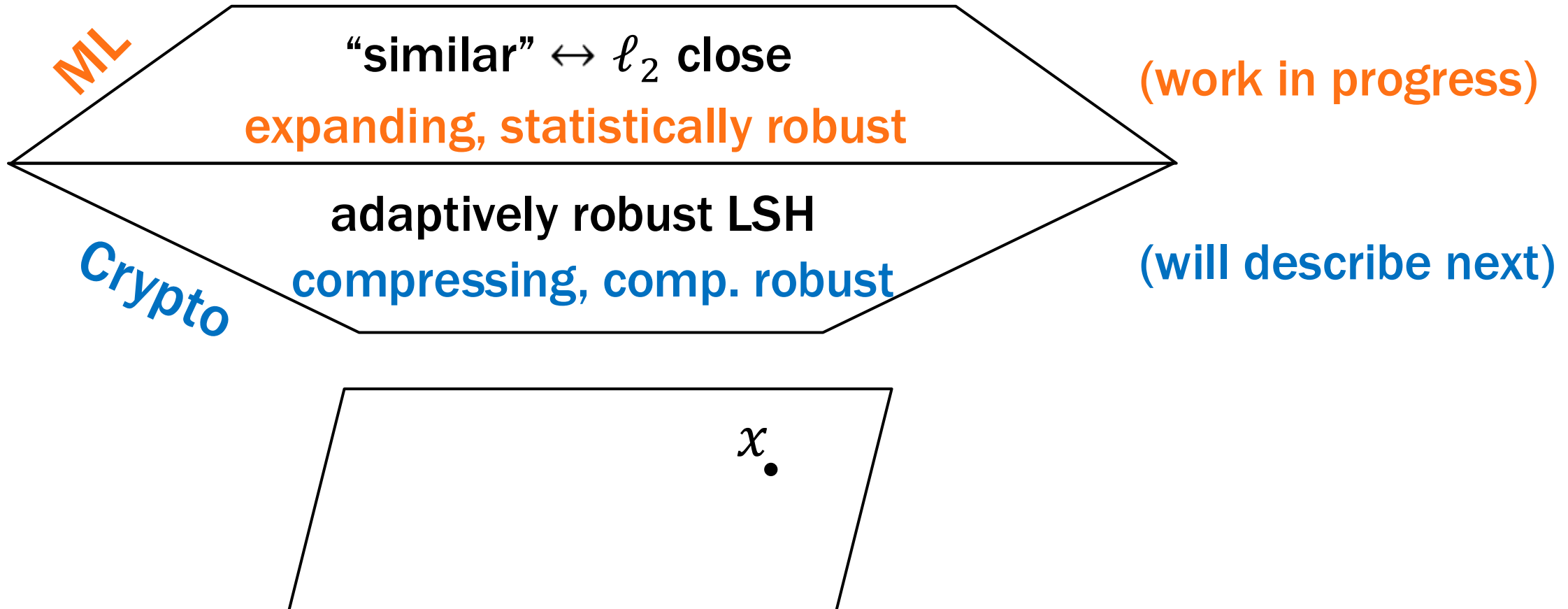
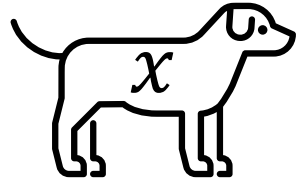
ML trained hash

ML Embeddings attempt to do two things in one shot:

1. ML magic: “similar” $\leftrightarrow \ell_2$ close
2. Compression

$x_{\bullet}$

New Paradigm: ML + Crypto





Robust Hamming LSH

Theorem [Boyle-Lavigne-V.'19, Holmgren-Liu-Tyner-Wichs'22]. Under the assumption that collision-resistant hash functions exist, there is a robust Hamming LSH. That is, for some n, d, ε , no adversary can produce $x, x' \in \{0,1\}^n$ violating the following conditions:

- if $\|x - x'\|_0 \geq d(1 + \varepsilon)$, then $\|h(x) - h(x')\|_0 \geq \tau$.
- if $\|x - x'\|_0 < d(1 - \varepsilon)$, then $\|h(x) - h(x')\|_0 < \tau$.



Robust Hamming LSH

Theorem [Boyle-Lavigne-V.'19, Holmgren-Liu-Tyner-Wichs'22]. Under the assumption that collision-resistant hash functions exist, there is a robust Hamming LSH (omitting parameters).

Starting point: if x, x' are close (resp. far), a random subset of coordinates of x and x' (of the right size) are likely to be equal (resp. different).

So, for “well-chosen” subsets $S_1, S_2, \dots, S_m$ for some m , the Hamming distance between $\mathbf{y} = (x|_{S_1}, \dots, x|_{S_m})$ and $\mathbf{y}' = (x'|_{S_1}, \dots, x'|_{S_m})$ “captures” the Hamming distance between x and x' .



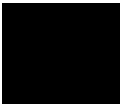
Robust Hamming LSH

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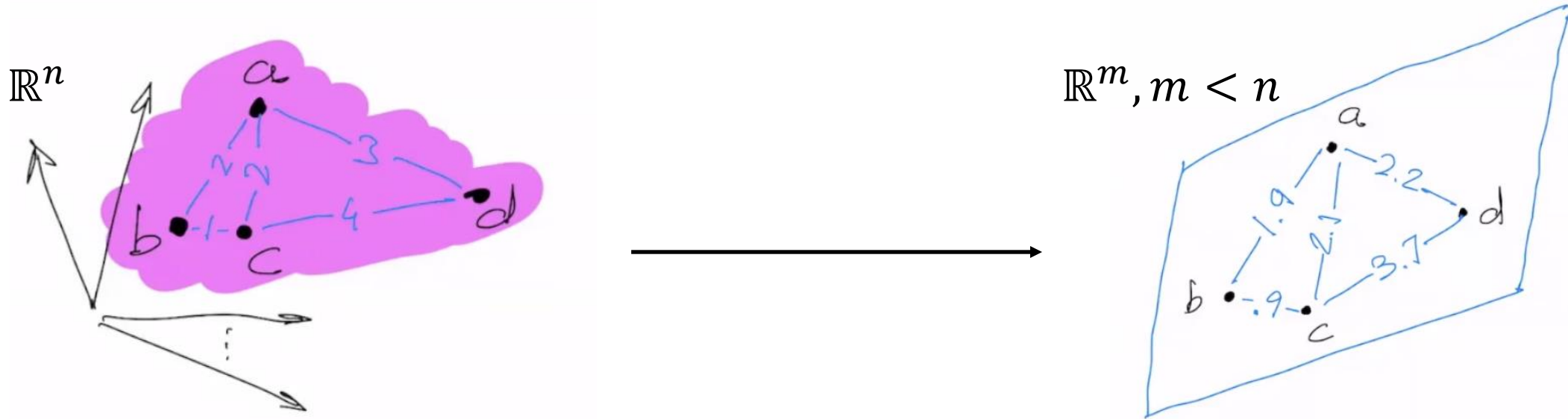
However, not hard to see that $\mathbf{y}$ is likely longer than x !

Solution: apply a collision-resistant hash to each $x|_{S_i}$.

$$\text{Ham-LSH}(x) = (\text{CRH}(x|_{S_1}), \text{CRH}(x|_{S_2}), \dots, \text{CRH}(x|_{S_m}))$$



Euclidean LSH



Lemma (Johnson-Lindenstrauss'84, Indyk-Motwani'99): Fix $0 < \varepsilon < 1$ and let $m = \Omega(\frac{\lambda}{\varepsilon^2})$. Let $\mathbf{h}(\mathbf{x}) = \mathbf{h}_A(\mathbf{x}) = \frac{1}{\sqrt{m}} \mathbf{A}\mathbf{x}$ where $A \sim N(0,1)^{m \times n}$ is a Gaussian matrix. For every $x, y \in \mathbb{R}^n$: $\leftarrow$ *chosen independent of A*

$$\Pr[||h(x) - h(y)|| \notin (1 \pm \varepsilon)||x - y||] \leq 2^{-\lambda}$$

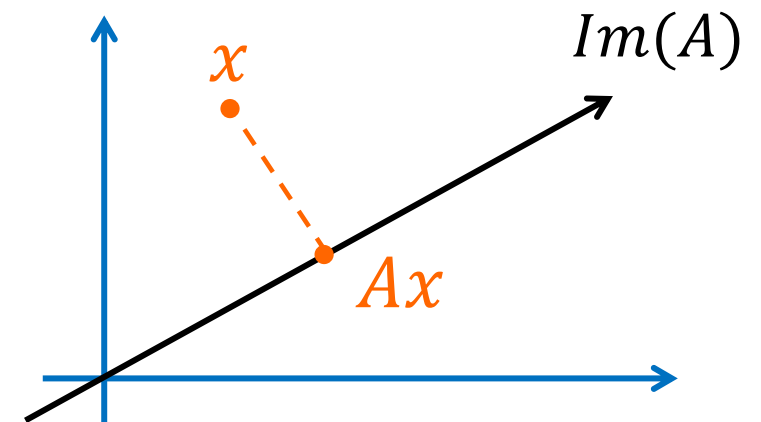
Robustness of JL?

Lemma (Johnson-Lindenstrauss'84, Indyk-Motwani'99): Fix $0 < \varepsilon < 1$ and let $m = \Omega(\frac{\lambda}{\varepsilon^2})$. Let $\mathbf{h}(\mathbf{x}) = \mathbf{h}_A(\mathbf{x}) = \frac{1}{\sqrt{m}} \mathbf{A}\mathbf{x}$ where $A \sim N(0,1)^{m \times n}$ is a Gaussian matrix. For every $x, y \in \mathbb{R}^n$: $\leftarrow$ *chosen independent of A*

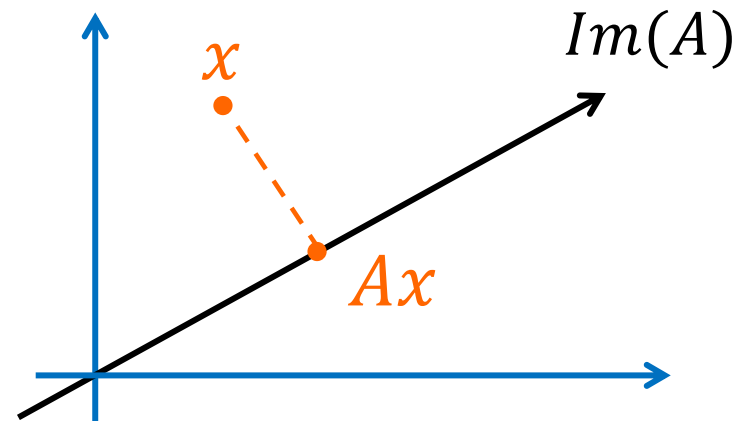
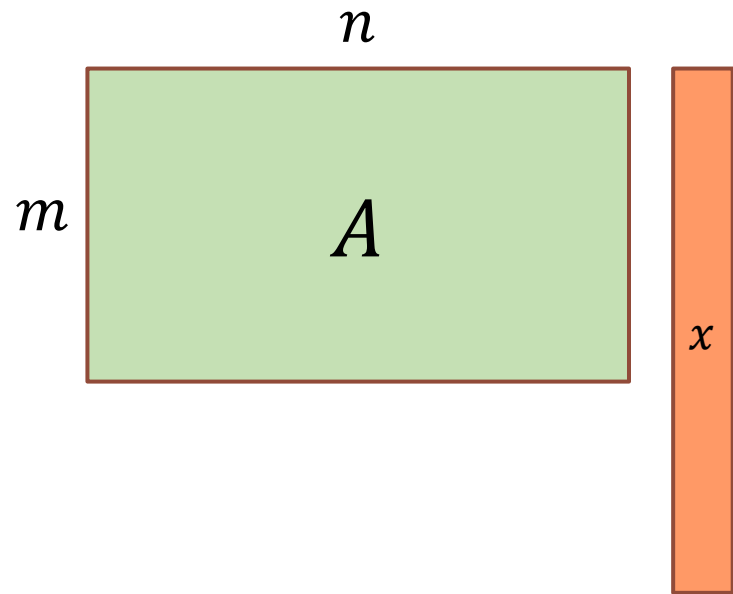
$$\Pr[||h(x) - h(y)|| \notin (1 \pm \varepsilon)||x - y||] \leq 2^{-\lambda}$$

Given A: easy! find a vector in the kernel of A.

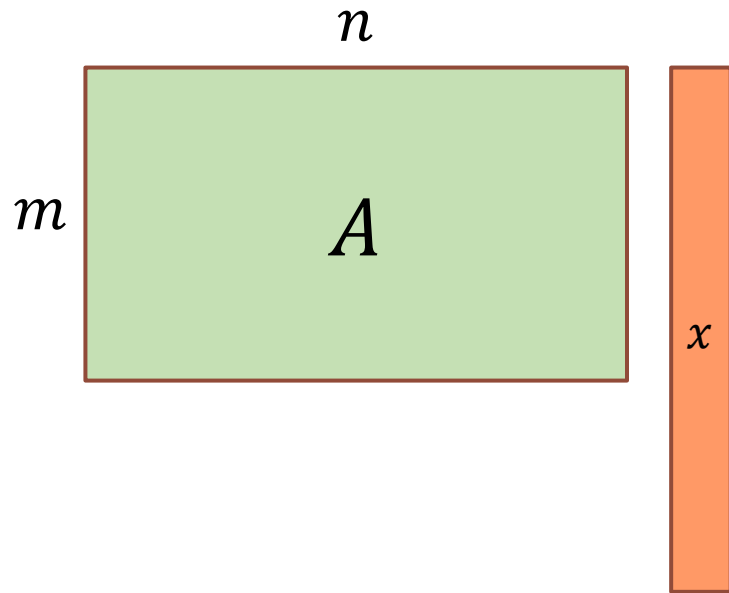
Even with super-weak “oracle access” to $\mathbf{h}_A$:
[Hardt-Woodruff'13] showed how to recover a “good enough” A' and run the kernel attack.



Robust Euclidean LSH: Hypergrid JL

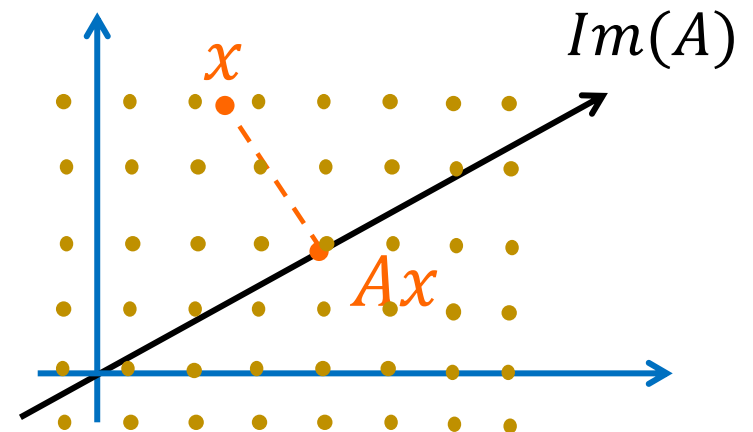
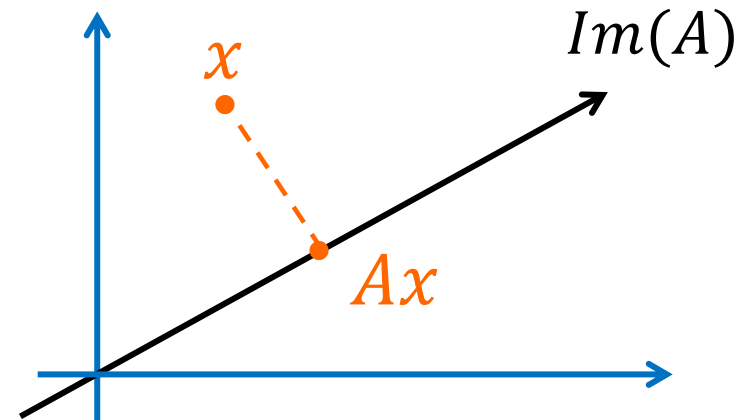


Robust Euclidean LSH: Hypergrid JL



x lives on the hypergrid $\{-b, \dots, b\}^n$.

Why the hypergrid? Practically motivated + kernel attack goes away.





Robust Euclidean LSH

Theorem [Bogdanov-Rosen-Vafa-V.'25]. Hypergrid JL gives a compressing, robust LSH for Euclidean distance over the hypergrid, with distortion bn/m , under the “contracting hypergrid vectors” (CHV) conjecture.

$$h_A(x) = \frac{1}{\sqrt{m}} \cdot Ax,$$

Expansion factor $\alpha \leq \sqrt{n/m}$ statistically, by spectral norm bounds on A .

Contraction factor $\beta \geq \frac{1}{b} \cdot \sqrt{\frac{m}{n}}$ under the “CHV conjecture”.

(more in Neekon's talk)

Put together, **Distortion** $\alpha/\beta \leq bn/m$.

“Real Cryptography”

Robust Euclidean LSH: natural problem

Needs computational assumptions.

Our bread-and-butter assumptions (even lattices) do not suffice.

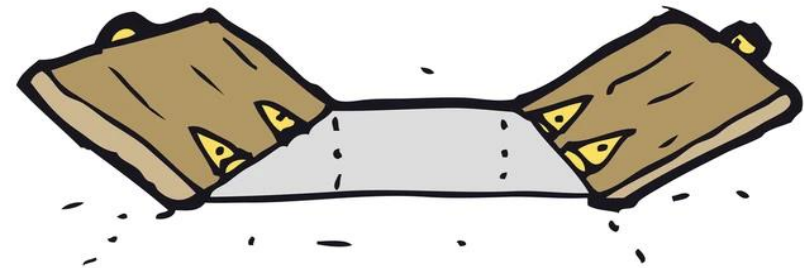
Contrast with Robust Hamming LSH for which CRHFs suffice [BLV'19, FS'21, FLS'22, Holmgren-Liu-Tyner-Wichs'22]

We need to use an assumption over $\mathbb{R}$: contracting hypergrid vectors.



Trapdoored Matrices & Applications

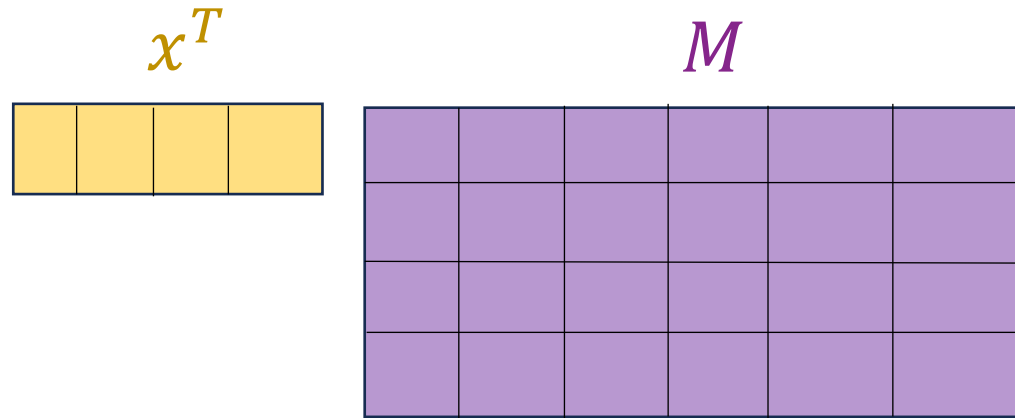
$$\begin{bmatrix} 1 & 2 & 2 & 4 & 5 \\ 2 & 4 & 6 & 8 & 10 \\ 3 & 7 & 9 & 13 & 15 \\ 4 & 8 & 11 & 16 & 20 \\ 5 & 10 & 15 & 20 & 23 \end{bmatrix}$$



Matrix Multiplication

Foundational to modern ML.

Consumes a significant fraction of compute cycles in training and inference.



Example: Embeddings (used for dim. reduction / clustering / ...)

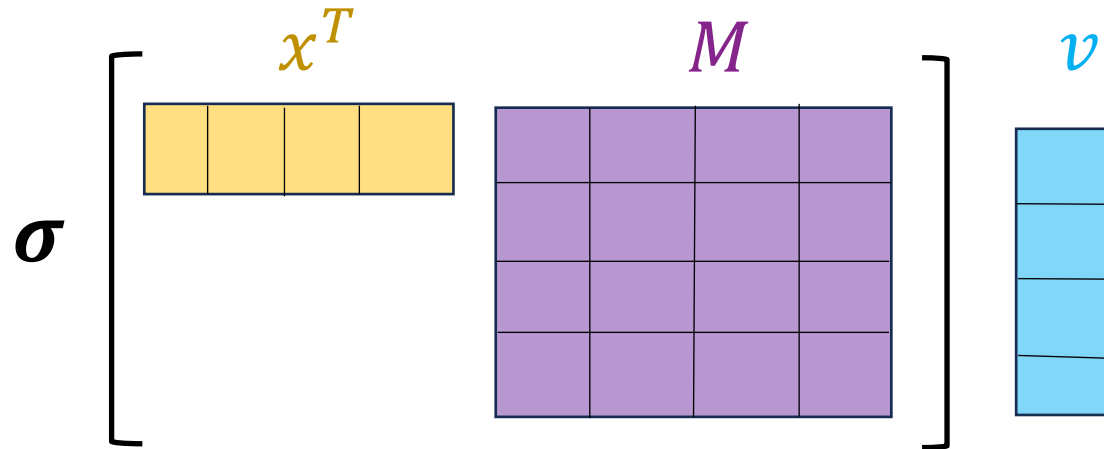
x : input, M : $n \times m$ random, $m > n$, (e.g. Gaussian)

Can we speed it up?

Matrix Multiplication

Foundational to modern ML.

Consumes a significant fraction of compute cycles in training and inference.



Example: Two-layer “random fratures” network

x : input, M : $n \times n$ random (Gaussian), v : “trained” weight vector

Can we speed it up?

Can we Speed it Up?

One can try to speed up matrix multiplication for specific applications.

e.g. “sparse Johnson-Lindenstrauss” method of Ailon and Chazelle 2009.

However: each application needs to be re-evaluated for correctness using potentially structured matrices.

e.g. “sparse subspace embeddings”, Chenakkod-Derezinski-Dong 2026.

A key novelty in our analysis is a matrix concentration technique we call *iterative decoupling*, which we use to fine-tune the higher-order trace moment bounds attainable via existing random matrix universality tools [Brailovskaya and van Handel, GAFA 2024].

WANT: a general-purpose speedup for *all (polynomial-time)* applications.



Trapdoored Matrices

Can you multiply a random $n \times n$ matrix M by a vector v in $\mathbf{o}(n^2)$ time ?

Trapdoored Matrices (V.-Zamir'25, Braverman-Newman'25): A pair (M, C) with $M \in \mathbb{F}^{n \times n}$ and C being a circuit over $\mathbb{F}$ where:

- M is ***computationally indistinguishable*** from random.
- C is a small, linear or near-linear size, circuit s.t. $\forall v: C(v) = Mv$.

Immediate Consequence: *Any* algorithm that uses multiplication of a vector by a random $m \times n$ matrix can be sped up by a factor of $\min(m, n)$ ***without degrading its quality / guarantees at all.***

Construction Idea (over finite fields)

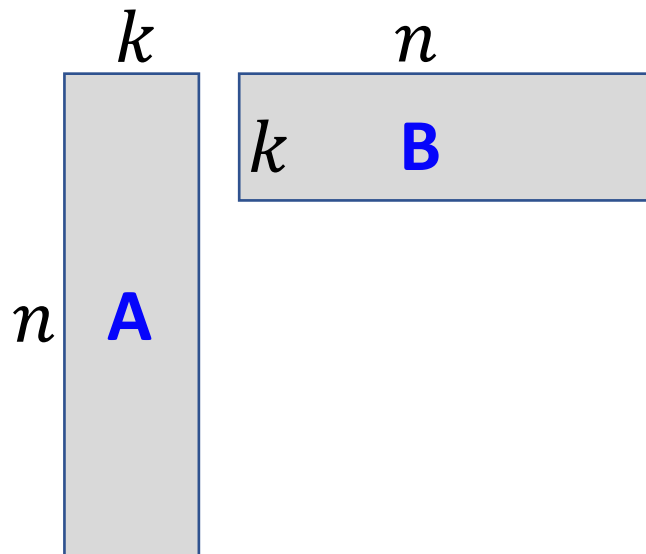
Which matrices admit linear or near-linear time matrix-vector multiplication?

These do not look “uniformly random” in any sense...

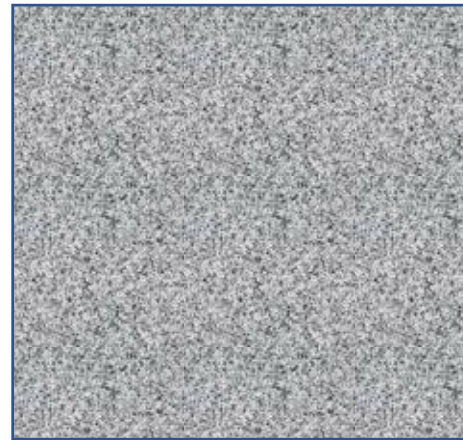
... but sums/products of these sometimes do.

Low-rank matrices

$$M = AB$$



Sparse matrices



“FFT” matrices

$$\begin{bmatrix} 1 & 1 & 1 & \dots & 1 \\ 1 & \omega & \omega^2 & \cdot & \omega^{n-1} \\ 1 & \omega^2 & \omega^4 & \cdot & \omega^{2(n-1)} \\ \dots & \cdot & \cdot & \dots & \dots \\ 1 & \omega^{n-1} & \omega^{2(n-1)} & \dots & \omega^{(n-1)^2} \end{bmatrix}$$

Construction Idea (over finite fields)

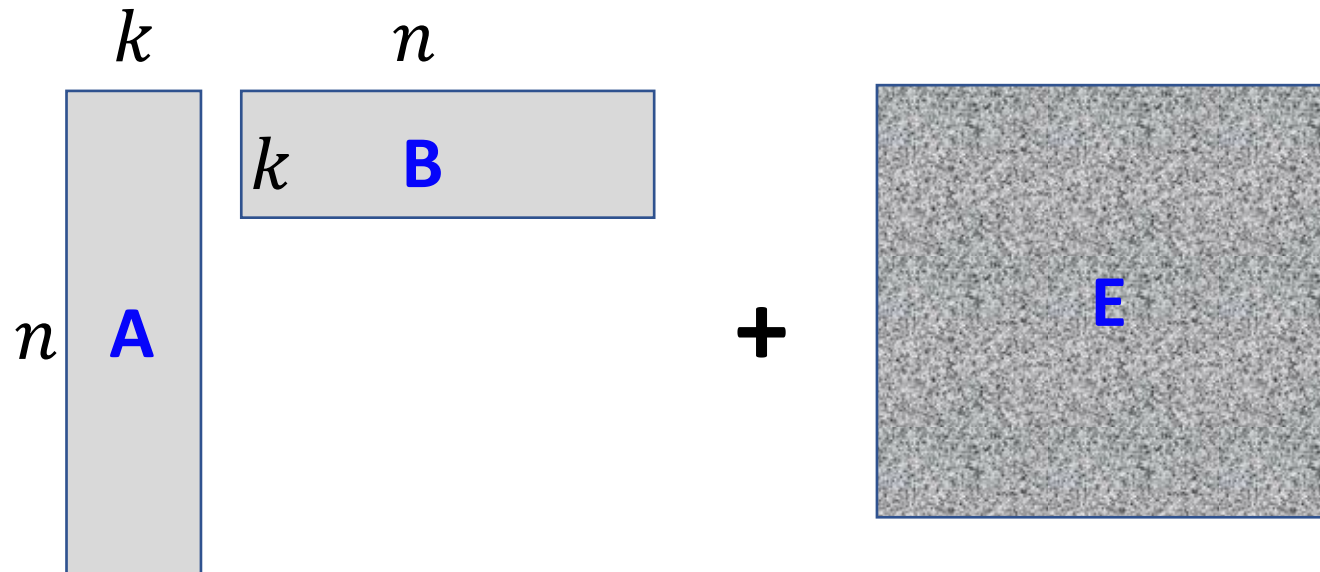
Which matrices admit linear or near-linear time matrix-vector multiplication?

These do not look “uniformly random” in any sense...

... but sums/products of these sometimes do.

e.g. The Learning Parity with Noise Assumption:

Sparse + Low-rank computationally indistinguishable from random



Construction over $\mathbb{R}$: the Kac Walk

Kac's Walk: What can we say about products of t random elementary rotation matrices in n dim?



$$R_{i,j,\theta}(e_k) = e_k$$

$$R_{i,j,\theta}(e_i) = \cos \theta \cdot e_i + \sin \theta \cdot e_j$$

$$R_{i,j,\theta}(e_j) = -\sin \theta \cdot e_i + \cos \theta \cdot e_j .$$

The Kac (Random) Walk: Nodes are matrices, edges are defined by the transitions above.

Stationary distribution: “uniform distribution” (Haar-measure) over $SO(n)$.

Construction over $\mathbb{R}$: the Kac Walk

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Stationary distribution: “uniform distribution” (Haar-measure) over $SO(n)$.

Conjecture: Kac's walk is “pseudorandom” after $\tilde{O}(n)$ steps.

If yes, we get a trapdoor matrices over $\mathbb{R}$ (where indistinguishability is w.r.t. the Haar measure or, with some work, any Haar-invariant measure).

Construction over $\mathbb{R}$: the Kac Walk

Fast mixing: Let Q_t be the distribution of matrices after t time steps.

- [Ailon-Chazelle'06] conjecture that Q_t for $t = \Omega(n \log n)$ is “random enough” for the JL transform; [JPSSS'20] proved this conjecture.
- [Jan01, MM13, PS17] show that **each column** of Q_t is TV-close to uniform on the sphere after **$t = \Omega(n \log n)$** steps.
- [PS18, PS25] show that the mixing time for the entire Q_t is between $\Omega(n^2)$ and $O(n^3 \log n)$. **But that's too slow for us.**
- [Rivest-Sotaraki'15] showed how to use the pseudo-randomness of the Kac walk (over a finite field) for fine-grained key exchange.

Open: Evidence for the Kac walk conjecture or a disproof of it.

Open Problems

Where the notion of trapdoored matrices do not apply (out of the box, at least): all matrices are “learned” e.g. in the attention computation in transformers.

$$\text{Attn}_{W_K, W_Q, W_V}(X) = \sigma((XW_K)(XW_Q)^T)(XW_V)$$

“trained” matrices:

we don’t have complete control over them (but we do have some).

open: can you train trapdoored matrices?

Open Problems

How else can we use Crypto to *speed up* Algorithms?

we know well how to use crypto to reduce randomness, space, interaction.

Open Problems

Other Applications of Trapdoored Matrices?

[Braverman-Newman'25]: *secure* outsourced matrix operations.

I want you to compute Av_i for me, but I do not want reveal A to you.
Solution idea: reveal $A + M$ where M is a trapdoored matrix.

[Chen-Ishai-Mour-Rosen'25]: secret-key private information retrieval.

a complexity-theoretic application: “best-possible” one-query random self-reduction for matrix mult, improving [Hirahara-Shimizu'25a'25b] but conditionally (on LPN).

Open Problems

Other, Better, Constructions of Trapdoored Matrices?

[Benhamouda-Chen-Halevi-Krawczyk-Ishai-Mour-Rabin-Rosen'25]: improved construction from the “learning subspaces with noise” assumption

Other Trapdoored Objects?

trapdoored degree- d , n -variate polynomials with $O(nd)$ time evaluation?

Summary

AI is here.

Impressive progress, but many challenges: trustworthy AI?

Crypto can make progress on solving them

Different models, Different goals, Different adversaries

Need new ideas, new tools, new sources of hardness

Many open problems!

*Hardness meets deep nets
Backdoors, filters, trapdoored mats
Crypto shapes (trustworthy) AI.*

ChatGPT 5.1 Thinking

Thanks!



**...ASK WHAT
CRYPTO CAN
DO FOR AI**

